

A Methodology for Generating Synthetic Personas via LLMs for Data-Restricted Domains

*Validation Case: Early Detection of Online Grooming

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Abstract—The advancement of Natural Language Processing (NLP) systems in sensitive domains is frequently hindered by the inaccessibility of real-world data, which is protected by legal and privacy regulations. This paper proposes a general methodology for generating synthetic *personas* via Large Language Models (LLMs) to mitigate data scarcity. The early detection of online grooming serves as the validation case. The proposed architecture employs a dual calibration strategy: utilizing the OSAEBA report solely for demographic and contextual framing of the Brazilian landscape, and leveraging the PAN 2012 collection strictly as training data for linguistic calibration and *baseline* evaluation. A multi-agent ecosystem was developed to produce fictitious dialogues, which were then aggregated and represented via TF-IDF to capture lexical features across early-message cutoffs. Binary classification experiments show high internal separability of the synthetic *corpus* ($F_{0.5}$ of 1.00 in *leave-one-out* validation over 41 conversations), a result we attribute to structural artifacts of the persona construction rather than to fine-grained grooming sensitivity. However, cross-domain evaluation reveals a severe failure in domain transfer (*domain shift*) for the classical sparse models, directly explained by a null Jaccard coefficient indicating an absolute lack of lexical overlap. Furthermore, the observed instability in data augmentation underscores the challenges of integrating disparate representations. The study concludes that this approach facilitates the controlled investigation of simulated patterns while exposing the empirical limitations of static datasets in representing contemporary threats, all while upholding strict ethical rigor by avoiding the use of real sensitive data.

Index Terms—Synthetic Personas, LLM, Early Detection, TF-IDF, Natural Language Processing, Cross-Domain Evaluation.

I. INTRODUCTION

The automated monitoring and classification of online interactions inherently depend on massive volumes of data for the effective training of Machine Learning models. However, in sensitive domains—such as online child exploitation, online grooming [1], [2], and harassment—the acquisition of real textual data encounters severe ethical, legal, and privacy barriers [3]. Consequently, the development of early detection and mitigation strategies often relies on outdated datasets primarily restricted to the English language [4], hindering the innovation and applicability of localized Natural Language Processing (NLP) tools.

To address this gap, this paper introduces a structured methodology for generating synthetic textual data using Large

Language Models (LLMs). While data simulation via LLMs has gained traction [5], [6], the present approach introduces dynamic interactions based on highly restricted and psychologically calibrated fictitious *personas*. This modeling seeks to mitigate generation drift and ensure that the artificial dialogues reflect specific sociodemographic vulnerabilities, thereby supporting text mining efforts without compromising the security of real individuals.

As a validation case, we adopt the scenario of early detection of online grooming. This domain poses significant challenges for NLP systems due to the subtlety of its rhetorical progression, which encompasses the initial approach, deceptive trust-building, and psychological isolation [7], [8]. Detecting these patterns requires conversation-level aggregation and rapid inference within the very first exchanged messages.

The technical objective of this work focuses on a core challenge in Text Mining: the modeling and classification of structured conversations operating strictly within highly restricted data matrices. Differentiating from sociological analyses, this study evaluates the viability of early detection by systematically applying early-message cutoffs to truncate conversations. Through this framework, we meticulously investigate the lexical limitations, the representativeness of TF-IDF feature extraction, and the transferability of sparse linear classifiers (*LinearSVC* and *MultinomialNB*). By training on the classic PAN 2012 benchmark and evaluating on the generated synthetic slices, we conduct a detailed analysis of class imbalance and cross-domain evaluation (*domain shift*). To ensure reproducibility and scientific transparency, the source code, the orchestrating *pipeline*, and the *persona* modeling are fully available in a public repository¹.

II. RELATED WORK

The foundational work on the automatic detection of online grooming consolidated around the PAN 2012 competition [4]. This *benchmark* advanced the field by providing anonymized real chat logs, spurring the development of multi-step architectures [9] and conversation-level restriction strategies [10]. Pan-

¹<https://github.com/Maximiliansabino/personas-sinteticas-validacao>

zariello [11] extended the state-of-the-art on the PAN dataset specifically for the *early detection* scenario, demonstrating the efficacy of applying TF-IDF representations combined with linear classifiers over temporal dialog slices (i.e., early-message cutoffs).

Subsequently, deep neural methods (such as convolutional networks and *Transformer*-based models like BERT [12], [13]) have sought to improve predictive precision. Nevertheless, these models often encounter practical deployment obstacles, including high computational inference costs and decisional opacity [14]. Crucially, a central aggravating factor persists: the PAN 2012 *corpus* reflects lexical idiosyncrasies (e.g., outdated IRC *slangs*) from over a decade ago, exclusively in English.

Concurrently, artificial data synthesis has emerged as a viable paradigm for low-resource NLP tasks. Authors such as Moller et al. [5] and West et al. [6] have demonstrated the capacity of LLMs to distill knowledge and support *data augmentation*. However, the static generation of samples via unified *prompts* often fails to model the discursive friction inherent to grooming interactions. The present work addresses this gap by deploying autonomous agents grounded in the conceptual frameworks of O’Connell [1] and Olson et al. [7], thereby simulating the complex interactional dynamics essential for robust text mining algorithms.

III. METHODOLOGY

The methodology was conceived as a continuous flow of simulation, from the parameterized structuring of the agents to the classic preprocessing required for text mining.

A. Construction and Dual Calibration of Personas

The architecture rests on three types of fictitious *personas*: predatory, victims (targets), and neutral. Each entity is specified via a JSON file that orchestrates its linguistic characteristics (vocabulary and dialect), intentions, and approach phases (e.g., approach, isolation). To guide the data generation without violating the privacy of real victims, a dual calibration was adopted:

- **Contextual Calibration:** The OSAEBA repository [15], a Brazilian statistical *survey* focused on the digital behavior of youth, was utilized. The report provided the demographic framing, vulnerability levels, and platform contexts for the simulation. Importantly, OSAEBA was used solely as contextual and demographic evidence, and not as a textual or chat corpus.
- **Linguistic Calibration:** The original PAN 2012 *corpus* [16] served strictly as training and baseline data. It guided the architectural calibration and informed the vocabulary restrictions of the predatory *persona* for the English-language component, functioning as the ground truth for our *baseline* classification.

Generalization. The reusable core of the method is domain-agnostic: (i) dual calibration that separates contextual/demographic grounding from linguistic grounding, (ii) a provider-agnostic multi-agent orchestrator, (iii) export to a

benchmark-compatible schema, and (iv) a truncation-plus-TF-IDF evaluation pipeline. Porting the method to another data-restricted domain requires substituting only the calibration sources and the persona role definitions; the grooming case instantiates these components but does not constrain them.

B. Multi-Agent Architecture and Synthetic Pipeline

The final iterations employed configurable models in a provider/model format, combining `groq/llama-3.3-70b-versatile` and `anthropic/claude-haiku-4-5-20251001` across different roles (predators, victims, and neutral conversations). This matrix permitted the variation of LLM providers and conversational roles without requiring modifications to the core orchestrator.

The orchestrator mediated speaking turns, persisted the serialized history in MongoDB Atlas, and, upon completion, exported the *logs* into an XML format structurally identical to the PAN 2012 *schema*. This design choice ensured immediate compatibility and consumption by our established NLP *pipelines*.

C. Textual Preprocessing and Representation

Within the text mining framework, the early detection challenge requires the XML *logs* to be aggregated by conversation identifier and iteratively truncated at the first N messages ($N \in \{5, 10, 15, 20, 24, 50\}$). To prevent vector contamination in the synthetic instances, any interactions unilaterally interrupted by the safety *guardrails* of the LLM APIs were systematically excluded. This ensured the strict preservation of intact conversational samples, which were then segmented into canonical binary labels (1 for predatory behavior, 0 for normal conversations). After excluding guardrail-interrupted executions, the resulting synthetic evaluation corpus comprised 41 intact conversations (21 predatory, 20 neutral), a deliberately small set whose size bounds the external validity of all synthetic-only results.

Feature extraction from the standardized text (tokenized and lowercased) was conducted using a strict TF-IDF model. To capture the combined slang structures and deceptive tactical sequences inherent to the predatory approach, the extraction incorporated both unigrams and bigrams (`ngram_range=(1, 2)`). The computational dimensional space was strictly constrained to the top 50,000 most frequent document descriptors (`max_features=50_000`); this design constraint effectively purges artifacts, isolated interjections, and the highly noisy *long-tail* typical of non-normative chat environments. Finally, to mitigate the exacerbated weight of linear superabundance (e.g., intentional *floods* or repetitive text anomalies), sublinear scaling was applied to the term frequencies (`sublinear_tf=True`). This classical formulation provides high algorithmic explainability for text mining, though it demands technical scrutiny regarding its sensitivity to multilingual and open-domain morphosyntactic variations.

IV. EXPERIMENTAL DESIGN

The design sought to answer the question of efficiency and lexical transferability. In real operational scenarios for network detection, the saturation of false positives induces systematic fatigue in evaluators. Consequently, $F_{0.5}$ (F-Score with weighting $\beta = 0.5$) was selected as the fundamental guiding metric. Unlike the generalist F1 score, the $F_{0.5}$ index penalizes false positives more vigorously, strongly prioritizing detection precision without completely discounting recall, which was tracked secondarily [11].

Given the sparse matrices originating from the restrictive lexical representation, we prioritized the non-probabilistic base hyperplane of *LinearSVC* (C=1.0, max_iter=10_000, random_state=42), establishing it as a solid and direct linear *baseline* for the classification of highly dimensional text vectors. Complementarily, *Multinomial Naive Bayes* (*MultinomialNB*; alpha=0.1) was included to support analytical comparisons as a traditional, strictly probabilistic alternative for conversational NLP. Because the primary intent is empirical demonstration and direct comparative reproduction across *datasets*, we deliberately abstained from systematic hyperparameter *fine-tuning*.

The PAN 2012 repository presents a scenario of severe native class imbalance, registering an asymmetry of approximately 10:1 (an overwhelming majority of normal users). All internal validations were anchored using robust stratified splits via *StratifiedKFold* (5 folds). It is imperative to note that random undersampling, when applied, was strictly isolated to the training fold to categorically prevent any *data leakage* into the testing nodes. Table I details the specific parameterizations of the mining pipeline.

Table I
TECHNICAL PARAMETERS OF THE MINING PIPELINE

Component	Value	Technical Justification	Implication/Limitation
Truncations	$N \in \{5..50\}$	Scale temporal detection restriction.	Severely reduces the global panorama focusing on initial contact.
TF-IDF	ngram_range=(1,2)	Protect two-word approach tactics.	Heightened sensitivity to pure slang and regional dialects.
Vocabulary	max_features=50,000	Purge long graphical distortions from the chat tail.	Chronic loss of the presence of extremely rare expression tokens.
Frequency	sublinear_tf=True	Dampen the impact of repetitive byte floods.	Prevents mathematical sequential spams from distorting distances.
LinearSVC	C=1.0	Configure non-flexibilized separating line.	Absence of grid search penalizes the optimization ceiling for comparative robustness.
Validation	Stratified K-Fold (5)	Protect the balance of proportions in 10:1 samples.	Folds with a sparse matrix may retain peripheral biases.
Sampling	Undersampling	Reduce the dominant influence of normality.	Shrinks absolute representativeness in the final training block.

Table II
INVESTIGATIVE PROTOCOL OF THE EXPERIMENTS (E1–E5)

Exp	Evaluation	Technical Investigative Question of Text Mining
E1	Baseline	Do LinearSVC and MultinomialNB algorithms strictly reflect classic reference results (PAN) in non-manipulated regimes?
E2	Transfer	Does the sparse threshold trained on the PAN dialect transfer predictive detection to samples of artificial national dialect?
E3	Consistency	Through <i>Leave-One-Out</i> with 95% CI via <i>bootstrap</i> , is it verified if the synthetic samples are internally discriminable?
E4	Diagnostics	Restricting to the top-20 TF-IDF feature selector, is mutual exclusion in the Jaccard representation attestable?
E5	Augmentation	Does the attachment and integration of simulated material dimensionally perturb an originally established massive model?

V. RESULTS

A. E1: Inherent Effects and the Detriment of Undersampling

The *baseline* (E1) confirmed the viability of early classification on PAN 2012 under the unadjusted class-imbalance regime. The LinearSVC performance scaled linearly: achieving an $F_{0.5}$ of 0.8331 with the initial 5 messages, reaching 0.9545 ($N = 20$), and stabilizing at 0.9670 ($N = 50$). The probabilistic MultinomialNB classifier yielded similar metrics, operating near 0.9645 for larger message contexts (Figure 1).

Surprisingly, despite historical recommendations [11], the application of undersampling deteriorated the $F_{0.5}$ metric across all message cutoffs in our configuration. In the restricted $N = 20$ cutoff, the undersampled LinearSVC was limited to an $F_{0.5}$ of 0.7394, heavily penalized by a drastic drop in precision. Technically, by discarding the vast majority of the neutral interactions that define the global vector separation boundary, the models forced a spurious increase in recall at the cost of an excessive rate of false positives. This contrast with the favorable undersampling results reported by Panzariello [11] is explained by our evaluation protocol: undersampling was confined to the training fold while the test fold preserved the natural $\approx 10:1$ distribution, so the precision cost of a balanced decision boundary is fully exposed rather than masked by a balanced test set.

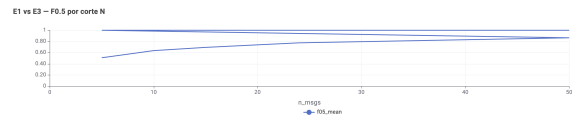


Figure 1. Evaluation of the *baseline* (E1) on PAN 2012 compared to the *Leave-One-Out* validation (E3) per truncation N .

B. E2 and E4: The Lexical Barrier and Explicit Negative Transfer

In the cross-domain evaluation (E2), predictive performance degraded severely. The LinearSVC model trained exclusively

on the PAN dataset did not transfer to the synthetic instances: although precision remained high at most cutoffs, recall collapsed to near zero (e.g., 0.05–0.14), driving $F_{0.5}$ down to 0.34 at intermediate truncations and to 0.00 at the most informative cutoff ($N = 50$), where the few positive predictions were also incorrect.

Experiment 4 (Jaccard) exposed the algorithmic root of this domain shift failure. By extracting and intersecting the top-20 *unigrams* and *bigrams* carrying the highest TF-IDF weights in each *corpus*, the calculated Jaccard coefficient was rigorously **0.00** (Figure 2). This confirms an absolute lack of lexical overlap. This explicit negative result indicates that the lexical dependence of sparse vector models severely impairs transferability between the English IRC slang of PAN 2012 and the Luso-Brazilian syntax generated by the synthetic *personas*. Together, E2 and E4 provide strong evidence of the empirical limits of static repositories as *proxies* for modeling localized and contemporary linguistic contexts.



Figure 2. The Jaccard coefficient on TF-IDF *features* provided evidence of complete lexical divergence between PAN and the artificial *corpus* (E4).

C. E3: Internal Consistency and Discriminability

To verify whether the cross-domain transfer failure in E2 originated from domain disparities rather than random noise in the generation process, a *Leave-One-Out* evaluation was strictly applied to the synthetic set to assess its internal consistency. In the final round, the *Leave-One-Out* validation on the synthetic corpus achieved an $F_{0.5} = 1.0000$ across all evaluated cutoffs ($N \in \{5, 10, 15, 20, 24, 50\}$), with precision, recall, and the 95% *bootstrap* CI (1,000 resamplings) identically at 1.0000 over 41 samples. This indicates a high degree of internal discriminability within the synthetic corpus. However, it is crucial to emphasize that this result does not imply external generalization, nor does it substitute for validation against independent corpora. Furthermore, the perfect separability observed as early as $N = 5$, combined with the null Jaccard overlap in E4, suggests that the model is likely detecting structural artifacts inherent to the persona construction itself, rather than exhibiting fine-grained sensitivity for early grooming detection.

D. E5: Instability in Data Augmentation

The attempt to integrate the two repositories via data augmentation yielded explicitly unstable and inconclusive results. The performance *delta* (the difference in percentage points when injecting artificial data into the PAN training matrix) oscillated radically depending on the size of the stochastic partition. It fluctuated from decreases in specific cutoffs (e.g., −0.45 p.p. at $N = 50$) to marginal artificial gains in the stratified matrix, as illustrated in Figure 3. Statistically, the

extreme volumetric discrepancy between the thousands of PAN conversations (6,450 training examples in the E5 80/20 stratified *holdout*) and the small representativeness of the synthetic volume simply diluted the semantic influence of the generated data within the global TF-IDF vector space.



Figure 3. Inconclusive deviation of $F_{0.5}$ after data augmentation (E5), indicating the dilution of the semantic influence of the synthetic corpus against the original volume.

VI. DISCUSSION AND ERROR ANALYSIS

The collective analytical evaluation of the results provides a classic illustration of the systematic barrier inherent to *domain shift* and the biases induced in text mining without deep semantic mapping. The drastic limitations observed in E2, aligned with the purely null intersection in the restricted positional Jaccard TF-IDF calculation (E4), explicitly signal the profound vocabulary dependence of *bag-of-words* approaches. These methods struggle significantly when faced with simultaneous cross-linguistic lags (American *slang* versus the Brazilian Portuguese syntax and *prompts* utilized by the *personas*). The resulting model collapse is indicative of an implicit global *overfitting* to the restricted temporal and lexical domain of the source database.

Regarding generated reliability, E3 provides evidence that the subcontexts generated by the LLM API maintain very clean internal discriminability boundaries, yielding ceiling scores in parametric sample isolation. However, from a generalized search and text mining perspective, this highlights intra-system coherence rather than demonstrating predictive efficacy for open, real-time external applications, especially given the small size of the currently generated output cluster. Moreover, the dilution observed when injecting the sparse synthetic vectors into the massive *full_train* matrix underscores the immense noise introduced by under-representation during the impure statistical augmentation of partitions in E5. This reveals the practical complications of merging minuscule and linguistically heterogeneous samples into an architecture that relies on mass co-occurrence to establish robust linear decision boundaries. Consequently, extreme caution is advised when performing lexical modeling on LLM-generated instances using strictly sparse TF-IDF extractions.

VII. LIMITATIONS AND ETHICAL CONSIDERATIONS

All dialogues generated during the execution phase are entirely fictitious. The system pragmatically adhered to the safety guidelines (*guardrails*) inherent to the utilized LLM APIs. Consequently, this iteratively truncated several executions during the initial approach phase due to automated blocks against synthetic sexual exploitation. The experimental design explicitly opted against the coercive deactivation of

these safety mechanisms, prioritizing ethical conduct over artificially inflating the total volume of conversations available for the mining phase. No generated sample exhibits literal dynamics or the intrinsic explicit terminology of sensitive material. Furthermore, this study did not involve the collection of new personal data or the exposure of identifiable real conversations; it strictly utilized fictitious synthetic data and the preexisting PAN 2012 database, which is anonymized and licensed for research purposes. Any formal review by an Ethics Committee should follow institutional norms, ensuring adherence to best practices for research involving secondary databases and simulated data.

VIII. CONCLUSION

The experiments indicate that the generated corpus exhibits high internal consistency within the scope of TF-IDF *features*. However, the restrictive experimental design highlighted empirical limitations, explicitly demonstrating cross-domain transfer failures. These failures stem from the strong local lexical dependence required by sparse vector methods when confronted with temporal and linguistic domain shifts. We suggest, therefore, that the evolution of these synthetic methodologies should encompass expanded corpora and explore textual representations less reliant on graphical term counts, with the goal of achieving more robust generalizations in future applications.

DECLARATION OF AI USE

It is declared, for technical-scientific transparency, that generative artificial intelligence tools were used to support programming, evidence organization, textual review, and editorial preparation. The methodological definition, data curation, interpretation of results, validation of claims, and final responsibility for the content remained exclusively with the human authors.

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